

Complex conjugate pairs

Basic skills

Q1. a) $z = 5 + i$
 $z^* = 5 - i$ ↘ change sign of imaginary part

b) $z = 6 - 2i$
 $z^* = 6 + 2i$

c) $z = 2(3i - 1) = 6i - 2$
 $z^* = -6i - 2$

Q2. a) $z^2 + 4z + 5 = 0$

$$z = \frac{-4 \pm \sqrt{16 - 20}}{2} = \frac{-4 \pm \sqrt{-4}}{2}$$

$$= \frac{-4 \pm 2i}{2} = -2 \pm i$$

$$z = -2 + i \text{ or } z = -2 - i$$

b) $z^2 + 9 = 0$ $z^2 = -9$

So $z = \pm \sqrt{-9} = \pm 3i$

$z = 3i$ or $z = -3i$

c) $z^2 - 2z + 5 = 0$

$$z = \frac{2 \pm \sqrt{4 - 20}}{2} = \frac{2 \pm \sqrt{-16}}{2}$$

$$= \frac{2 \pm 4i}{2} = 1 \pm 2i$$

$z = 1 + 2i$ or $z = 1 - 2i$

Competency skills

Q3. a) $2z^2 + z + 5 = 0$

$$z = \frac{-1 \pm \sqrt{1 - 40}}{4} = \frac{-1 \pm \sqrt{-39}}{4}$$

b) $4z^2 = -16$ $z^2 = -4$ $z = \pm \sqrt{-4}$

So $z = \pm 2i$

$$c) \quad 8z + 5(1-z^2) = 0$$

$$\Rightarrow 5z^2 - 8z - 5 = 0$$

$$z = \frac{8 \pm \sqrt{64 - 100}}{10} = \frac{8 \pm \sqrt{-36}}{10}$$

$$= \frac{8 \pm 6i}{10} = \frac{4 \pm 3i}{5}$$

$$d) \quad 2z = \frac{z^2}{4} + 25$$

$$\Rightarrow z^2 - 8z + 100 = 0$$

$$z = \frac{8 \pm \sqrt{64 - 400}}{2} = \frac{8 \pm \sqrt{-336}}{2}$$

$$z = \frac{8 \pm 4i\sqrt{21}}{2} = 4 \pm 2\sqrt{21}i$$

Q4. If $z = 3 + 4i$ solves it then
so does the conjugate $z^* = 3 - 4i$

$$\text{so } (z - (3+4i))(z - (3-4i))$$

$$= z^2 - \underbrace{(3+4i)z - (3-4i)z}_{\text{notice imaginary cancels}} + \underbrace{(3+4i)(3-4i)}_{w \times w^*}$$

$$= z^2 - 6z + 9 + 16$$

$$= |w|^2$$

$$= z^2 - 6z + 25$$

Q5. If $z = 5 - i$ is a solution, then
so is the conjugate $z^* = 5 + i$

$$\text{so } (z - (5-i))(z - (5+i))$$

$$= z^2 - (5-i)z - (5+i)z + (5-i)(5+i)$$

$$= z^2 - 10z + 25 + 1$$

$$= z^2 - 10z + 26$$

Advanced skills

$$\text{Q6. } z^3 - 5z^2 + z + 75$$

$$\text{sols: } z = 4 - 3i \text{ so also}$$

$$z = 4 + 3i$$

Then factors $(z - (4 - 3i))(z - (4 + 3i))$

$$= z^2 - 8z + 16 + 9 = z^2 - 8z + 25$$

so $z^3 - 5z^2 + z + 75 = (z + a)(z^2 - 8z + 25)$

a must be 3

so $(z + 3)$ is other factor

so $z = -3$ is third solution

Q7.

$$z^3 - 5z^2 + 9z - 5$$

has $(z - i + 2) = (z - (i - 2))$

as a factor then

$(z - (-i - 2))$ is also a factor

hence

$$(z + a)(z - (i - 2))(z - (-i - 2))$$

$$(z + a)(z^2 + 4z + 5)$$

$$= z^3 - 5z^2 + 9z - 5$$

$$\text{So } a = -1$$

hence Fully factorised

$$z^3 - 5z^2 + 9z - 5 = (z-1)(z-i+2)(z+i+2)$$

Q8. a) $z = 4 + 2i$

$$z^* = 4 - 2i \quad zz^* = (4 - 2i)(4 + 2i)$$

$$zz^* = 4^2 + 2^2 = 20$$

b) $z = 8 - 3i$

$$zz^* = 8^2 + 3^2 = 64 + 9 = 73$$

Q9. $z = x + yi$

$$\bar{z} = x - yi$$

$$z\bar{z} = (x + yi)(x - yi)$$

$$= x^2 + xyi - xyi - y^2i^2$$

$$= x^2 + y^2$$

Q10. For $az^2 + bz + c = 0$

$$z = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

For complex solutions $b^2 - 4ac < 0$

$$\text{SO } z = \frac{-b \pm i\sqrt{4ac - b^2}}{2a}$$

as $\pm$ imaginary part we have conjugates

This reasoning can be extended to cubics as 1 real + $\begin{cases} 2 \text{ real} \\ 1 \text{ real} \\ 2 \text{ complex} \end{cases}$

quartics as $\begin{cases} 2 \text{ real} \\ 1 \text{ real} \\ 2 \text{ complex} \end{cases} + \begin{cases} 2 \text{ real} \\ 1 \text{ real} \\ 2 \text{ complex} \end{cases}$
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